# 9 Applied Statistics



Walter Shewhart
(March 18, 1891-
March 11, 1967)

## Introduction

The term Applied Statistics refers to the use of statistical theory to conduct operational activities in a variety of fields in real life situations. Today, the applications of statistics are an indispensable part of any and every activity. There are many fields in which statistical concepts can be applied, some of them are business decision making, finance, marketing, economics, social sciences, industry, agriculture etc… An important aspect of applied statistics is to study about the present and future behaviour of the activities performed in an industry.

Walter Andrew Shewhart (pronounced like "shoe-heart",) was an American physicist, engineer and statistician, sometimes known as the father of statistical quality control and also related to the Shewhart cycle.

### Learning Objectives

After studying this chapter the students are able to understand

- Time series data
- Components of Time Series
- Moving Averages
- Seasonal Variation
- Index Numbers
- Weighted Index Number
- Tests for an Ideal Index Number
- Statistical Quality Control
- Causes for Variation
- Process Control and Product Control

YP1GD5

In this chapter we would be studying about the theoretical and application of the statistical methods of Time Series, Index Number and Statistical Quality Control. Each one of them has its importance in its field of application. Statistical analysis has been widely used for



XII Std - Business Maths & Stat EM Chapter 9.indd 200 2/28/2020 5:11:01 PM